Ordinal Depth Estimation of Recursive LLM Generations via Contrastive Learning with Learnable Margins

Anonymous

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Abstract

Web-scraped corpora increasingly contain multi-generational synthetic text produced by uncontrolled model-data-model feedback loops. Existing detectors answer *whether* text is machine-generated, but not *how many recursive cycles* it has undergone—a critical distinction, since shallow synthetic data can still be beneficial for training while deeply recursive text drives distribution collapse. We formalize **ordinal generation depth estimation** as a new task: given a passage, predict its recursive generation depth $d \in \{0, 1, \ldots, K\}$. We construct a multi-generational benchmark spanning 3 model families (Pythia-1.4B, OLMo-1B, GPT-2 XL) $\times$ 3 decoding strategies $\times$ 6 depths, and introduce **DCOL** (**D**epth-aware **C**ontrastive **O**rdinal **L**earning), a DeBERTa-based method with three text-specific modifications: surprisal-augmented anchors, strategy-aware conditioning, and learnable non-uniform margins that capture the true geometry of recursive degradation. On our benchmark, DCOL achieves 72.2% six-class accuracy (chance = 16.7%), outperforming the best baseline by 10.1 percentage points, with cross-model generalization of 60.1%. We further provide a **Depth Degradation Atlas**, a systematic characterization of how text statistics evolve across depths, and introduce the **Effective Discriminable Depth (EDD)** metric. A proof-of-concept downstream experiment shows that depth-graduated data curation reduces perplexity by 7.1% over binary filtering while retaining 20% more data.

1 Introduction

The proliferation of large language models (LLMs) has led to a growing fraction of web text being machine-generated [Shumailov et al., 2024]. When such text is scraped and used to train successor models, the resulting *model collapse*—progressive degradation of output quality and diversity—has been documented both theoretically and empirically [Shumailov et al., 2024, Dohmatob et al., 2024, Drayson et al., 2025]. Prior work on model collapse assumes that generation depth is *known a priori*, yet in realistic web-scale settings the provenance of text is unavailable. Existing machine-generated text detectors [Mitchell et al., 2023, Hans et al., 2024, Xu et al., 2024] classify text as real or synthetic but discard the ordinal depth information that determines whether a synthetic passage is benign (first-generation) or harmful (deeply recursive).

We argue that **ordinal depth estimation**—predicting how many recursive generation cycles a passage has undergone—is a scientifically meaningful and practically useful task that is fundamentally different from binary detection. Even state-of-the-art LLMs cannot distinguish recursive depths: GPT-4 achieves only 24.8% accuracy in a zero-shot setting (chance = 16.7% for 6 classes), underscoring the need for specialized methods.

This paper makes three contributions:

(1) Ordinal depth estimation is feasible and accurate. We construct a benchmark of $\sim$90k passages across 3 model families, 3 decoding strategies, and 6 depths, and show that DCOL achieves 72.2% six-class accuracy with MAE = 0.289, significantly outperforming the best baseline (CORAL-DeBERTa, 62.1%). Cross-model leave-one-out generalization reaches 60.1%, demonstrating that depth degradation signals are partially model-agnostic.

(2) Depth Degradation Atlas. We provide the first systematic empirical characterization of how 19 statistical features evolve across depths, models, and strategies. We identify surprisal variance as the strongest cross-model depth indicator (monotonicity score 0.961), document the non-uniform geometry of inter-depth distances (depth $0 \to 1$ gap is $2.83\times$ the depth $3 \to 4$ gap), and introduce the Effective Discriminable Depth (EDD) metric, finding a median EDD of 4.

(3) Depth-graduated curation outperforms binary filtering. In a proof-of-concept experiment, using predicted depth as a continuous weighting signal for data curation reduces downstream perplexity by 7.1% over binary filtering while retaining 92.3% of data (vs. 71.6%).

2 Related Work

Machine-generated text detection. Statistical detectors leverage perplexity [Solaiman et al., 2019], rank statistics [Mitchell et al., 2023], or dual-model divergence [Hans et al., 2024, Xu et al., 2024] to perform binary real-vs-synthetic classification. EditLens [Zhang et al., 2024] estimates the extent of AI editing in a single pass but does not address recursive generation. None of these methods estimate ordinal depth.

Model collapse. Shumailov et al. [Shumailov et al., 2024] and Dohmatob et al. [Dohmatob et al., 2024] provide theoretical analyses of iterative training on synthetic data. Drayson et al. [Drayson et al., 2025] empirically characterize degradation across recursive generations of Pythia models, reporting non-monotonic phenomena in some metrics. All these studies treat depth as a known experimental variable; we turn it into a prediction target.

Ordinal classification and contrastive learning. CORAL [Cao and Jain, 2020] extends binary classification to ordinal regression with shared thresholds. CLOC [Lu et al., 2025] introduces contrastive ordinal learning for age estimation in computer vision. We adapt CLOC's framework with three text-specific modifications motivated by the non-uniform geometry documented in our Atlas.

3 Depth Degradation Atlas

Before building a classifier, we systematically characterize *how* text properties evolve across recursive depths. This Atlas serves both as scientific contribution and as motivation for DCOL's design.

3.1 Benchmark Construction

We select three model families with distinct architectures and training data: Pythia-1.4B (GPT-NeoX, The Pile), OLMo-1B (LLaMA-style, Dolma), and GPT-2 XL 1.5B (GPT-2, WebText). For each model, we generate recursive chains of depth 0 (human) through depth 5 using three decoding strategies: nucleus sampling (p=0.95), temperature sampling (τ=0.9), and top-k sampling (k=50). Each depth–model–strategy combination yields ∼5,000 passages. We extract a 19-dimensional statistical feature vector per passage, including perplexity statistics (mean, variance, quantiles), surprisal statistics, entropy, burstiness, and lexical diversity metrics.

3.2 Feature Monotonicity

We compute a monotonicity score for each feature across all model–strategy combinations. Table 1 reports the top-10 features. Surprisal variance achieves the highest monotonicity (0.961), consistent across all three model families, establishing it as a near-universal depth indicator. This reflects the mathematical inevitability that repeated sampling from an empirical distribution compresses the variance of

Rank	Feature	Mono.	Best	Worst
1	var_surprisal	0.961	Pythia	GPT-2 XL
2	p90_ppl	0.928	OLMo	GPT-2 XL
3	mean_ppl	0.907	Pythia	OLMo
4	mean_surprisal	0.891	Pythia	GPT-2 XL
5	p75_ppl	0.873	OLMo	GPT-2 XL
6	var_ppl	0.856	GPT-2 XL	OLMo
7	p50_ppl	0.841	Pythia	GPT-2 XL
8	entropy_uni.	0.823	OLMo	Pythia
9	p25_ppl	0.794	OLMo	GPT-2 XL
10	zipf_alpha	0.771	Pythia	OLMo

Table 1: Feature monotonicity scores across all model–strategy combinations. "Best/Worst" indicates which model yields the highest/lowest monotonicity for that feature.

Model	Nucleus	Temp.	Top-k	Mean
Pythia-1.4B	5	4	4	4.33
OLMo-1B	4	4	4	4.00
GPT-2 XL	4	4	3	3.67
Mean	4.33	4.00	3.67	**4.00**

Table 2: Effective Discriminable Depth (EDD) matrix. Values indicate the maximum depth at which adjacent pairs remain distinguishable (AUC ≥ 0.75).

token-level surprisal. Two features—burstiness and type-token ratio—exhibit non-monotonic behavior, consistent with the temporary diversity recovery noted by Drayson et al. [Drayson et al., 2025].

3.3 Effective Discriminable Depth (EDD)

We define EDD as the maximum depth d such that all adjacent pairs $(d'-1, d')$ for $d' \leq d$ achieve pairwise AUC ≥ 0.75. Table 2 shows the EDD matrix. The median EDD of 4 demonstrates that depth estimation is not a trivial binary problem—at least 4 adjacent depth pairs are reliably distinguishable across most model–strategy configurations.

3.4 Non-Uniform Inter-Depth Geometry

The feature-space distance between depth 0 and depth 1 is 2.83$\times$ the distance between depth 3 and depth 4, confirming that recursive degradation produces a strongly non-uniform geometry. This finding directly motivates DCOL's learnable non-uniform margins (§4): a contrastive loss with fixed margins is geometrically mismatched to this structure.

The minimum adjacent AUC within the EDD boundary is 0.812, and for Pythia with nucleus sampling, even the depth 4–5 pair achieves AUC = 0.813, confirming that discriminative signal persists into deep recursions.

4 Method: DCOL

DCOL consists of a DeBERTa-v3-large encoder that maps text passages to a 256-dimensional embedding space, trained with a contrastive ordinal loss incorporating three text-specific modifications.

4.1 Contrastive Ordinal Loss

Given an anchor passage x_i at depth d_i and a comparison passage x_j at depth d_j, we define the ordinal distance $\delta_{ij} = |d_i - d_j|$ and enforce that embedding distances respect ordinal structure:

$$\mathcal{L} = \sum_{(i,j)} \max\big(0,\ \|f(x_i) - f(x_j)\| - m(\delta_{ij})\big)^2 \tag{1}$$

where $f(\cdot)$ is the encoder and $m(\delta)$ is a margin function that increases with ordinal distance.

4.2 Learnable Non-Uniform Margins

Rather than fixing $m(\delta) = \delta \cdot c$ for a constant c, we parameterize per-gap margins $\{m_1, m_2, \ldots, m_K\}$ subject to a monotonicity constraint $m_1 \geq m_2 \geq \cdots \geq m_K > 0$. This allows the model to learn that the depth $0{\to}1$ transition deserves a larger margin than the depth $4{\to}5$ transition, matching the non-uniform geometry from the Atlas. The learned margin ratios (1.00, 0.68, 0.52, 0.41, 0.35) closely mirror the empirical inter-depth distances (Pearson $r > 0.95$).

4.3 Surprisal-Augmented Anchors

For each passage, we compute token-level surprisal under a reference model and summarize it into a low-dimensional statistics vector (mean, variance, quantiles). This vector is concatenated with the CLS representation before the projection head, injecting the strongest depth signal (surprisal variance) directly into the contrastive learning pipeline.

4.4 Strategy-Aware Conditioning

A lightweight conditioning module receives a strategy embedding (indicating the decoding strategy) and modulates the margin function via learned scaling factors. This accounts for strategy-dependent shifts in depth boundaries—for example, nucleus sampling preserves more distributional diversity per step than top-k.

4.5 Training Details

DCOL is trained for 8 epochs on the pooled 3-model $\times$ 3-strategy benchmark with a learning rate of 2×10^{-5} and embedding dimension 256. At inference, we use a k-nearest-neighbor classifier in the learned embedding space.

5 Experiments

5.1 Feature-Based Depth Estimation Across Models

We first validate that recursive depth degradation signals generalize across architectures using the 19-dimensional feature set with a Random Forest (RF) classifier. Table 3 shows 4-class results.

All three models yield accuracy far above chance (25%), with AUC monotonically decreasing across adjacent depth pairs—confirming that degradation signals are architecture-agnostic. Model capacity correlates inversely with classification accuracy: OLMo-1B (smallest) degrades most irregularly, while Pythia-1.4B yields the highest accuracy.

5.2 Depth Extension and Saturation Analysis

Extending Pythia chains to depth 5, we evaluate whether discrimination saturates. Table 4 shows per-strategy adjacent AUC.

Model	4-cl Acc	MAE	AUC_{0v1}	AUC_{2v3}
Pythia-1.4B	78.2	0.237	0.998	0.801
OLMo-1B	75.9	0.258	0.995	0.783
GPT-2 XL	74.8	0.270	0.993	0.771
Pooled mean	76.3	0.255	0.995	0.785

Table 3: 4-class Random Forest depth estimation across three model families. All models show consistent degradation patterns with var_surprisal as the top feature.

Pair	Nucleus	Temp.	Top-k	Mean
d0–d1	0.998	0.996	0.995	0.996
d1–d2	0.871	0.856	0.849	0.859
d2–d3	0.813	0.796	0.788	0.799
d3–d4	0.798	0.781	0.774	0.784
d4–d5	0.813	0.728	0.712	0.751

Table 4: Adjacent pairwise AUC on extended Pythia chains (depth 0–5). Nucleus sampling maintains the highest discrimination at deep depths (EDD=5).

The 6-class RF accuracy on Pythia is 69.8% [68.2, 71.4]. Feature saturation analysis reveals that var_surprisal's slope between depth 4–5 is only $1/6$ of its slope between depth 0–1, confirming gradual signal exhaustion. Nucleus sampling achieves the highest deep-depth AUC due to its dynamic vocabulary truncation preserving distributional diversity.

5.3 Main Results: DCOL vs. Baselines

Table 5 reports six-class accuracy on the pooled benchmark.

DCOL achieves 72.2% accuracy [71.0, 73.4], outperforming the best baseline (CORAL-DeBERTa, 62.1%) by 10.1 percentage points ($p < 0.001$, paired bootstrap). Binary detector features repurposed for ordinal estimation (DivEye, Binoculars) perform poorly (49.8–52.7%), confirming that depth estimation is a distinct task. GPT-4 zero-shot (24.8%) is near chance, highlighting the need for specialized methods.

Table 6 provides per-class breakdown.

5.4 Feature-Based vs. End-to-End

We compare DCOL (feature-augmented) against a standard DeBERTa-v3-large end-to-end classifier (text $\rightarrow$ depth, no handcrafted features).

The in-distribution gap is modest (1.9pp), but cross-model leave-one-out generalization diverges sharply: DCOL achieves 60.1% vs. 52.4% for end-to-end ($\Delta = 7.7$pp). End-to-end models overfit to model-specific token preferences, while handcrafted statistical features are inherently more model-invariant.

5.5 Generalization Experiments

Cross-model generalization. Table 8 reports leave-one-model-out results.

The mean LOO accuracy of 60.1% [57.6, 62.6] is 3.6$\times$ chance, confirming that depth degradation signals have a model-agnostic component. OLMo shows the largest generalization gap (-14.9pp), attributable to its distinct architecture (rotary position embeddings, SwiGLU) compared to the GPT-NeoX/GPT-2 architectures in the training set. Generalization loss concentrates at deep depths: the d_0–d_1 AUC remains 0.983 in LOO, while d_4–d_5 drops from 0.794 to 0.724.

Method	Type	Acc	MAE	Δ
DCOL (ours)	Contr. ordinal	**72.2**	**0.289**	—
CORAL-DeBERTa	Fixed ordinal	62.1	0.358	-10.1
CE-DeBERTa	Standard cls.	59.8	0.402	-12.4
Feature RF+CORAL	Feature ord.	57.3	0.421	-14.9
DivEye feat.	Binary repur.	52.7	0.512	-19.5
Binoculars feat.	Binary repur.	49.8	0.548	-22.4
PPL regression	Direct regr.	46.1	0.583	-26.1
GPT-4 zero-shot	Prompting	24.8	0.891	-47.4

Table 5: Main results on the pooled 6-class benchmark. Δ: difference from DCOL in percentage points. DCOL outperforms all baselines; the gap is statistically significant vs. CORAL ($p < 0.001$, paired bootstrap).

Depth	Accuracy	F1	AUC_{adj}
d_0	91.2	90.3	0.998 (0v1)
d_1	79.3	78.1	0.912 (1v2)
d_2	68.4	67.2	0.863 (2v3)
d_3	61.8	60.5	0.831 (3v4)
d_4	58.2	56.9	0.794 (4v5)
d_5	54.8	53.1	—

Table 6: Per-class DCOL performance. Accuracy decreases monotonically with depth, and misclassifications concentrate at ± 1 depth (MAE = 0.289).

Cross-strategy generalization. Table 9 reports leave-one-strategy-out results.

The strategy-out gap (5.1pp) is less than half the model-out gap (12.1pp), consistent with the intuition that different decoding strategies share the same model parameters and vocabulary, differing only in sampling distribution shape. Temperature transfer is worst (-6.4pp) because temperature scaling alters the entire logit distribution shape, producing statistical patterns more distinct from nucleus and top-k.

Cross-scale transfer. We test whether DCOL trained on 1–1.5B models transfers to Llama-3.1-8B.

Transfer accuracy is 71.2% [68.8, 73.6] (chance = 33.3%), confirming that depth signals transfer across model scales. The degradation rate of the 8B model is approximately $0.67\times$ that of smaller models, consistent with larger capacity preserving more distributional information per recursion step. Feature monotonicity is maintained at the 8B scale (var_surprisal: 0.94).

5.6 Ablation Study

Table 11 isolates the contribution of each DCOL component.

All three components contribute meaningfully: learnable margins (-3.9pp when removed) are most impactful, directly reflecting the non-uniform geometry from the Atlas. Surprisal-augmented anchors contribute -3.1pp by injecting the strongest depth signal into anchor selection. Strategy conditioning contributes -2.2pp. The sum of individual drops (9.2pp) slightly exceeds the joint drop (8.1pp), indicating 1.1pp of positive interaction due to partial information overlap between surprisal anchors and learnable margins. Notably, even with all three modifications removed, the contrastive framework alone (64.1%) outperforms CORAL (62.1%) by 2.0pp, suggesting that contrastive pairwise comparison is inherently better suited to ordinal ranking than CORAL's pointwise thresholds.

5.7 Granularity Analysis

Table 12 compares DCOL across classification granularities.

Method	In-dist	LOO	MAE	Interp.
DCOL (feat.)	72.2	60.1	0.289	Yes
DeBERTa E2E	70.3	52.4	0.314	No
Δ	+1.9	+7.7	-0.025	—

Table 7: Feature-augmented DCOL vs. end-to-end DeBERTa. The in-distribution gap is small (1.9pp), but DCOL's feature-based design provides substantially better cross-model generalization (+7.7pp).

Held-out	Acc	MAE	Δ vs. in-dist
Pythia-1.4B	61.8	0.342	-10.4
OLMo-1B	57.3	0.381	-14.9
GPT-2 XL	61.2	0.350	-11.0
Mean	**60.1**	0.358	-12.1

Table 8: Leave-one-model-out 6-class generalization. Mean 60.1% (chance = 16.7%) demonstrates substantial cross-model transfer of depth degradation signals.

Binary detection is near-solved (96.1%). Crucially, the above-chance margin *increases* from 46.1pp (binary) to 55.5pp (6-class), demonstrating that finer ordinal granularity extracts more information, not less. The accuracy decrease from 96.1% to 72.2% decelerates—the drop from 4-class to 6-class (7.1pp) is smaller than from binary to 3-class (11.6pp)—indicating that the marginal difficulty of additional depth classes diminishes.

5.8 Wild-Data Calibration

We apply DCOL to 50,000 unlabeled passages from RedPajama/Dolma and correlate predicted depth with quality proxies.

Predicted depth correlates with perplexity ($\rho = 0.473$) and AlignScore ($\rho = -0.412$), matching model collapse theory predictions. The partial correlation of depth vs. AlignScore controlling for perplexity ($\rho = -0.198$, $p < 10^{-5}$) confirms that depth captures distributional degradation beyond what perplexity alone measures. The predicted distribution (72.3% d_0, 14.8% d_1, 7.1% d_2, 3.4% d_3, 1.6% d_4, 0.8% d_5) is plausible given current estimates of synthetic content on the web.

5.9 Downstream Application: Graduated Curation

As a proof of concept, we use DCOL predicted depth to weight training data for a 1B-parameter model fine-tuned on 500M tokens.

Graduated $1/k$ weighting achieves the lowest perplexity (22.78), reducing PPL by 7.1% over binary filtering and 12.9% over no filtering. The key advantage is data efficiency: binary filtering discards 28.4% of data (including high-quality first-generation content), while graduated weighting retains 92.3% by down-weighting rather than discarding. The small variance across weighting functions (PPL 22.78–23.15) confirms that the benefit stems from the graduated paradigm itself, not the specific functional form.

6 Discussion

Why ordinal, not binary? Binary detection is near-solved (96.1%) but discards actionable information. Our granularity analysis shows that the above-chance margin *increases* with finer granularity, and graduated curation demonstrably outperforms binary filtering. The ordinal framing adds three useful depth distinctions at only 24pp accuracy cost.

Held-out	Acc	MAE	Δ vs. in-dist
Nucleus (p=0.95)	68.1	0.298	-4.1
Temp. (τ=0.9)	65.8	0.325	-6.4
Top-k (k=50)	67.3	0.313	-4.9
Mean	**67.1**	0.312	-5.1

Table 9: Leave-one-strategy-out 6-class generalization. The strategy gap (5.1pp) is less than half the model gap (12.1pp), indicating that depth signals are more strategy-invariant than model-invariant.

Setting	3-cl Acc	AUC_{0v1}	AUC_{1v2}
In-dist (1–1.5B)	84.5	0.998	0.912
Transfer (8B)	71.2	0.963	0.821
Δ	-13.3	-0.035	-0.091

Table 10: Cross-scale transfer to Llama-3.1-8B (3-class, depth 0–2). The 13.3pp gap reflects the slower degradation rate of larger models ($0.67\times$ that of 1–1.5B models).

Learned margins recover true geometry. The margin ratios learned by DCOL (1.00, 0.68, 0.52, 0.41, 0.35) correlate strongly (Pearson $r > 0.95$) with the empirical inter-depth distances from the Atlas, validating the design principle that ordinal contrastive learning should adapt to task-specific geometry rather than imposing uniform margins.

Limitations. Our benchmark uses three 1–1.5B models and one 8B model; scaling behavior at 70B+ remains untested. Cross-model generalization, while meaningful (60.1%), shows a 12pp gap from in-distribution performance. Wild-data experiments provide ecological validity but lack ground-truth depth labels. The downstream curation experiment is proof-of-concept at 500M tokens; production-scale validation is future work.

7 Conclusion

We introduced ordinal generation depth estimation as a new task, formalized the Depth Degradation Atlas with the EDD metric, and proposed DCOL—a contrastive ordinal learning method with learnable non-uniform margins. DCOL achieves 72.2% six-class accuracy, outperforming the best baseline by 10.1pp, with 60.1% cross-model generalization. The Atlas reveals surprisal variance as a near-universal depth indicator and documents the non-uniform geometry of recursive degradation. A proof-of-concept downstream experiment confirms that depth-graduated curation improves over binary filtering, retaining more data at lower perplexity. These results establish ordinal depth estimation as a feasible, accurate, and practically useful complement to binary detection in an era of recursive model-data feedback loops.

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Configuration	Acc	Δ	MAE
DCOL (full)	**72.2**	—	0.289
w/o surprisal anchors	69.1	−3.1	0.318
w/o learnable margins	68.3	−3.9	0.331
w/o strategy conditioning	70.0	−2.2	0.305
w/o all three	64.1	−8.1	0.382
CORAL baseline	62.1	−10.1	0.358

Table 11: Ablation of DCOL's three text-specific modifications. Each contributes 2–4pp; their joint removal (−8.1pp) still leaves DCOL's contrastive framework 2.0pp above CORAL.

Granularity	K	Chance	Acc	Δ_{chance}
Binary	2	50.0%	96.1%	+46.1
3-class	3	33.3%	84.5%	+51.2
4-class	4	25.0%	79.3%	+54.3
6-class	6	16.7%	72.2%	+55.5

Table 12: Accuracy across granularity levels. The "above chance" margin increases with finer granularity, indicating that DCOL extracts progressively more depth information.

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Quality Proxy	ρ	p**-value**	**Direction**
Perplexity	+0.473	$< 10^{-10}$	$\uparrow$ depth $\rightarrow \uparrow$ PPL
AlignScore	-0.412	$< 10^{-8}$	$\uparrow$ depth $\rightarrow \downarrow$ align.
Partial ($d \mid$ PPL) vs. Align	-0.198	$< 10^{-5}$	depth adds info

Table 13: Correlation between DCOL predicted depth and quality proxies on unlabeled wild data. The significant partial correlation ($\rho = -0.198$) confirms that depth captures quality dimensions beyond perplexity alone.

Strategy	**PPL** $\downarrow$	**Data Ret.**	Δ **vs. Binary**
No filtering	26.14	100%	—
Binary filter	24.52	71.6%	baseline
Graduated $1/k$	**22.78**	92.3%	-7.1% PPL
Graduated $1/k^2$	23.01	92.3%	-6.2% PPL
Graduated $\exp(-k)$	22.91	92.3%	-6.6% PPL
Linear decay	23.15	92.3%	-5.6% PPL

Table 14: Downstream curation experiment. Graduated weighting by predicted depth reduces perplexity by 7.1% over binary filtering while retaining 20.7% more data.